

Towards a Global Quantum Agenda

UNESCO Global Quantum Initiative for Sustainable Development (2026-2028)



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A UNESCO Initiative under the International Decade of Sciences for Sustainable Development

Executive Summary

The quantum revolution stands at a critical juncture. While global investments in quantum science and technology (QST) exceed \$55.7 billion and are projected to reach \$106 billion by 2040, a profound "quantum divide" is emerging between nations and within societies. Over 150 countries lack national quantum strategies, and participation from the Global South remains severely limited. Without coordinated action, quantum sciences and technologies risk becoming another source of global inequality rather than a tool for shared prosperity.

Gaps in research capacity, educational opportunities, infrastructure, and public awareness are widening, while international collaboration remains fragmented and global governance mechanisms underdeveloped. These asymmetries risk reinforcing existing scientific and technological inequalities, excluding much of the world from shaping and benefiting from the quantum transformation.

To prevent this emerging divide from solidifying into long-term structural inequities, the international community must act collectively and decisively. This calls for a shared global agenda rooted in cooperation, ethical foresight, and inclusive capacity development. Only through such coordinated efforts can quantum science become a force for sustainable development and shared global progress.

In this spirit, UNESCO, GESDA, ITU, ... [could founding partners be listed here] are launching a multi-year international initiative, the Global Quantum Initiative for Sustainable Development (GQI). This initiative aims to provide a collaborative platform through which Member States, scientific institutions, industry actors, and other stakeholders can co-develop strategic priorities, exchange knowledge, and jointly shape an inclusive and forward-looking global framework.

The proposed GQI deliberately balances ambition with pragmatism, **pursuing a vision for inclusive global leadership in quantum science and technology while ensuring organizational credibility through achievable milestones, transparent resource allocation, and measurable outcomes. It also creates a structured platform for stakeholders to determine collectively whether such a Global Quantum Agenda (GQA) is necessary, what value it would add, and how it should be shaped.**

Over a **three-year period (2026-2028)**, the GQI will implement pilot projects, facilitate regional and thematic consultations and build practical insights into quantum capacity needs, governance approaches and cooperation models leading to the development of the **GQA in 2030**. This approach allows for careful validation of concepts, gradual resource mobilization, and iterative refinement based on pilot experiences - thereby **avoiding the credibility risks** associated with overpromising and underdelivering. It also reflects lessons learnt from past peaks in scientific enthusiasm, such as graphene and other frontier materials, where disproportionate investment, misaligned expectations, and premature scaling led to stagnation rather than transformation. In contrast, the GQI places **inclusion, capacity building, and science diplomacy** at the forefront, ensuring that a potential global quantum agenda is grounded in real needs and practical experience.

Key features of the proposed framework include: a focus on **bridging the quantum divide** through targeted capacity building; establishment of sustainable cooperation mechanisms rather than one-time events; emphasis on South-South and North-South collaboration; integration of ethical governance from the outset; and pragmatic resource mobilisation aligned with the operational capacity of UNESCO and all of the founding partners of GQI. .

The total budget request of **\$3.6 million over three years** for GQI represents a strategic investment in UNESCO and other founding partners' future leadership in quantum governance, with clear deliverables, measurable outcomes, and built-in mechanisms for partner co-funding and resource multiplication.

1. Context and Strategic Rationale

1.1 The Quantum Landscape: Promise and Risks

Quantum science and technology represents one of the most significant scientific frontiers of the 21st century, with the potential to revolutionize computing, communications, sensing, and materials science. The global quantum technology market, currently valued at approximately \$1.88 billion in 2025, is experiencing unprecedented growth, with a 27.3% increase from the previous year. Major economies have committed substantial resources to quantum development, with the United States allocating over \$1.2 billion through the National Quantum Initiative, the European Union investing €7.2 billion in the Quantum Flagship program, and China reportedly investing over \$15 billion in quantum research and development.

However, this rapid advancement has created a stark clash in global quantum readiness. While a handful of developed nations race to achieve quantum power, the vast majority of countries - particularly in the Global South - lack the basic infrastructure, human capital, and policy frameworks necessary to participate meaningfully in the quantum economy. **UNESCO's preliminary global survey on quantum science and technology**, which has gathered over 590 responses across all regions, reveals significant disparities in quantum

readiness, with many countries expressing urgent needs for policy support, capacity building, and access to quantum infrastructure.

This emerging "**quantum divide**" poses profound implications for global equity, sustainable development, and international cooperation. Unlike previous technological revolutions that evolved gradually, quantum technologies exhibit characteristics that make exclusion particularly problematic: **extremely high knowledge and technological barriers, critical dependence on specialized infrastructure, very limited basic education coverage, and profound potential social impacts**. Without deliberate intervention, quantum development risks creating a new form of **technological colonialism**, where quantum-advanced nations dominate global innovation while quantum-excluded countries become permanent consumers of quantum technologies they cannot understand, control, or influence.

1.2 Learning from Past Peaks of Scientific Hype

Recent experiences in the scientific landscape offer important lessons about the risks of prematurely scaling up initiatives without a sufficiently grounded research or application base. The case of graphene, once hailed as a revolutionary material with vast potential across industries—from electronics to energy storage, illustrates how early scientific breakthroughs, while promising, can stall when met with premature expectations and disproportionate investments. Despite strong initial momentum and large-scale funding, the pathway from discovery to transformative technology proved more complex and slow-moving than anticipated.

Such examples share common characteristics: ambitious promises unaligned with technological readiness, insufficient support for fundamental science, fragmented research agendas, and disconnection between policy enthusiasm and long-term capacity building. These dynamics can lead to public and political disillusionment and ultimately undermine trust in the broader scientific enterprise.

For UNESCO, and in the context of the Global Quantum Agenda, these lessons are critical. The proposed framework is intentionally designed to avoid the pitfalls of hype cycles through:

- Realistic goal-setting aligned with scientific maturity and resource levels
- Phased implementation that allows for learning and course correction
- Transparent monitoring and evaluation mechanisms
- Commitment to inclusive multilateral collaboration over isolated, donor-driven pursuits
- Safeguards to ensure continuity and technical integrity over political cycles

By grounding the agenda in both scientific realism and strategic foresight, UNESCO can help ensure that global momentum around quantum science translates into long-term societal benefit—not just short-lived excitement.

1.3 UNESCO's Unique Value Proposition

UNESCO is uniquely positioned to lead the development of a sustainable quantum future through the Global Quantum Initiative – a strategic framework designed to ensure quantum technologies benefit humanity as a whole in a long term and to call urgent action to bridging the Divide.

As the United Nations specialized agency with a cross-cutting mandate in science, education, culture, and communication, UNESCO is uniquely positioned to foster balanced dialogue, promote responsible governance, and enable international cooperation in quantum development. UNESCO can address this topic holistically rather than through narrow technical or economic lenses. This comprehensive approach is essential for quantum technologies, which will impact not only scientific research and economic competitiveness but also education systems, cultural practices, and social structures.

As emphasized during the Global Ministerial Dialogue on Science Diplomacy held at UNESCO in March 2025, UNESCO leverages its interdisciplinary expertise, longstanding partnerships with Member States, and proven track record in areas such as AI ethics, open science, and technology governance, to advance responsible quantum development for sustainable and inclusive global benefit. This effort draws upon UNESCO's global convening power as well as its leadership in open science, AI ethics, STI policy, STEM education, and capacity building.

UNESCO's established relationships with Global South countries, built through decades of capacity building and technical cooperation, provide a foundation of trust that commercial entities and developed-country-led initiatives cannot replicate. The organization's experience in managing complex multilateral processes, from the development of AI ethics recommendations to the implementation of open science frameworks, demonstrates proven capability in facilitating consensus-building around emerging technologies.

Furthermore, UNESCO's recent leadership in AI ethics has established important precedents for responsible technology governance. The UNESCO Recommendation on the Ethics of Artificial Intelligence, adopted by 193 member states in 2021, provides a template for addressing the ethical dimensions of quantum technologies while they are still in early development stages. This proactive approach to ethical governance represents a significant competitive advantage over initiatives that treat ethics as an afterthought.

The organization's global network of National Commissions, 54 field offices, and Category 1 Institutes, Category 2 Centres, Chairs and Networks and all relevant partners and associated institutions provide unparalleled reach for implementing capacity building programmes on UNESCO's unique mandate in science, and facilitating international cooperation. Unlike purely academic or commercial networks, UNESCO's institutional infrastructure can support sustained, long-term engagement with diverse stakeholders across all regions.

UNESCO's Vision: *A world where quantum technologies are developed responsibly, shared equitably, and contribute to sustainable development for all nations and communities.*

2. The Global Quantum Initiative: Vision and Objectives

The Global Quantum Initiative (GQI) is a three-year multilateral process and framework for action launched by UNESCO. It aims to mobilise global stakeholders to shape the future of quantum technologies through extensive consultation, collaboration and pilot projects.

The Global Quantum Agenda (GQA) represents the core deliverable of this initiative: a strategic, living document framework with broad international consensus. This agenda will provide shared principles, priorities and an action roadmap for the development, governance and cooperation of quantum technologies worldwide.

2.1 Defining the Global Quantum Agenda

The Global Quantum Agenda is envisioned as a **strategic framework for international cooperation in quantum science and technology, designed to ensure that the benefits of quantum development are shared equitably while addressing the challenges posed by quantum technologies responsibly**. The GQA emphasizes the social, educational, and diplomatic dimensions of quantum development, and takes care to avoid the pitfalls of overhyped technology rollouts disconnected from foundational readiness.

The GQA is conceived not as a static document or one-time declaration, but as an evolving framework that adapts to technological developments, changing geopolitical circumstances, and emerging social needs. This adaptive approach reflects the inherent uncertainty in quantum technology development and the need for flexible governance mechanisms that can respond to unforeseen challenges and opportunities.

Central to the GQA is the principle of "quantum inclusion" - the idea that all countries and communities should have opportunities to participate in and benefit from quantum development according to their needs, capabilities, and priorities. **This does not mean that all countries must develop identical quantum capabilities, but rather that the global quantum ecosystem should be structured to prevent the emergence of permanent quantum "haves" and "have-nots."**

2.2 Core Objectives of the Global Quantum Initiative in Developing the Global Quantum Agenda

The GQI has one overarching mission: to facilitate the co-creation of the Global Quantum Agenda (GQA). This mission is pursued through **two primary objectives**: (i) catalyzing

immediate actions... and (ii) orchestrating the inclusive and consensus-driven development process of the GQA itself.

To advance these objectives, the Initiative proposes five strategic entry points that will guide international cooperation and policy dialogue. These entry points reflect both the urgent thematic challenges requiring collective attention and the enabling mechanisms essential for fostering an inclusive and sustainable quantum future.

Through this Initiative, UNESCO and the founding partners seek to mobilize diverse stakeholders, foster policy coherence, and provide a structured pathway toward the co-creation of a consensus-driven Global Quantum Agenda focused on equitable innovation and sustainable development.

The objectives of the foreseen Global Quantum Initiative are:

Objective 1: Advancing Quantum Education and Strengthening Institutional and Human Capacity

- 1.1 Advise on the integration of quantum science into national development strategies and research agendas, fostering interdisciplinary research capacity and strengthening links between academia, industry, and policy institutions.
- 1.2 Advocate for the accelerated development of advanced research programmes, specialized training, and peer-learning platforms to build institutional capacity for quantum research, innovation, and technology transfer, leveraging international cooperation and regional research hubs
- 1.3 Promote the development of inclusive, multilingual, and open-access advanced learning and training resources, tailored to researchers, educators, and public sector professionals, to support capacity building in quantum science and its applications across diverse contexts.

The primary objective focuses on reducing disparities in quantum readiness between and within countries. This involves developing targeted capacity building programs that address the specific needs of different regions and stakeholder groups, with particular emphasis on strengthening capabilities in countries with limited quantum infrastructure. Capacity building encompasses multiple levels: individual (training scientists, policymakers, and educators), institutional (strengthening research institutions and policy frameworks), and systemic (developing national quantum strategies and international cooperation mechanisms).

The approach recognizes that capacity building cannot follow a one-size-fits-all model. Different countries have different starting points, priorities, and constraints. Some may focus on quantum education and awareness-raising, others on developing specific quantum applications relevant to their development needs, and still others on participating in global quantum research networks. The GQI framework provides advice on flexible pathways for countries to engage according to their circumstances while ensuring that no country is permanently excluded from quantum development opportunities.

Objective 2: Building Sustainable Quantum Infrastructure and Enabling Resource Sharing

- 2.1 Advocate for enabling national environments and institutional frameworks that support long-term investment in quantum research and innovation infrastructure.
- 2.2 Promote equitable access to shared facilities and quantum cloud services through international cooperation, regional platforms, and open-access initiatives.
- 2.3 Strengthen downstream innovation ecosystems and application pathways by facilitating the development of incubators, testbeds, and collaborative hubs that accelerate responsible deployment.

The second objective focuses on the practical infrastructure needed to support quantum development, including **research facilities, computational resources, and data sharing platforms**. Given the high costs and technical complexity of quantum infrastructure, sharing mechanisms are essential for ensuring that countries with limited resources can still participate in quantum research and development.

This objective includes encouraging the development of **remote access** programs that allow researchers in resource-limited settings to use quantum computers and other specialized equipment located elsewhere. It also encompasses the development of open-source quantum software and educational resources that can be freely accessed and adapted by users worldwide. The emphasis on sustainability ensures that infrastructure investments create lasting benefits rather than short-term capabilities that cannot be maintained.

Objective 3: Advancing Public Awareness and Societal Engagement

- 3.1 Promote inclusive campaigns to enhance quantum science literacy at all levels, fostering public understanding through accessible, culturally relevant content.
- 3.2 Encourage science communicators and trusted information channels to promote accurate, engaging narratives about quantum science to reach diverse audiences.
- 3.2 Facilitate participatory public engagement platforms and initiatives to involve youth, women and girls, educators, and communities in co-shaping the quantum future.

The third objective addresses the critical need for **quantum literacy** across different levels of society. This includes developing quantum education extra curricula for schools and universities, training teachers and educators, creating public engagement programmes to increase awareness of quantum technologies, and supporting the development of quantum-related skills in the workforce.

Quantum education during the foundational educational level presents unique challenges due to the counterintuitive nature of quantum phenomena and the mathematical sophistication required for deep understanding. The GQI framework emphasizes the development of approaches that make quantum concepts accessible to diverse audiences while maintaining scientific accuracy. This includes supporting the development of

educational resources in multiple languages and cultural contexts targeting school teachers and schools students, aligning with UNESCO's STEM programme, recognizing that effective quantum education must be adapted to local circumstances and learning traditions.

Objective 4: Strengthen International Cooperation through Science Diplomacy

4.1 Facilitate inclusive participation of Global South countries in international quantum dialogues, and agenda-setting processes through policy support and capacity-building programmes.

4.2 Promote inclusive, multi-stakeholder platforms to enable sustained dialogue and knowledge exchange among researchers, policymakers, industry stakeholders, and civil society across regions and sectors.

4.3 Foster the development of cooperation frameworks and mechanisms to advance responsible quantum technology transfer, joint research, and regional collaboration in support of equitable innovation, , to ensure Quantum Science and Technology for Sustainable Development.

The fourth objective recognizes that quantum development is inherently international in scope, requiring cooperation across borders, sectors, and disciplines. The GQA framework aims **to strengthen existing cooperation mechanisms while creating new platforms for collaboration that are more inclusive and representative than current arrangements.**

Science diplomacy plays a crucial role in this objective, as quantum technologies have significant implications for national security, economic competitiveness, and international relations. The GQA framework provides neutral ground for countries to engage in quantum cooperation even when broader political relationships may be strained. This diplomatic dimension is particularly important given the dual-use nature of many quantum technologies and the need to balance openness in scientific cooperation with legitimate security concerns.

Objective 5: Fostering Responsible Quantum Innovation and Governance

5.1 Promote inclusive and anticipatory governance frameworks that embed ethical, legal, and societal considerations into quantum research and innovation processes.

5.2 Encourage shared accountability among governments, scientists, and private actors by advancing global dialogues on responsibility, transparency, and equity in quantum development.

5.3 Advise on mechanisms for engagement in established international standardization processes to facilitate contributions to the development of international norms and technical standards to ensure safety, interoperability, and public trust in emerging quantum technologies.

The fifth objective addresses the need for **ethical frameworks** and governance mechanisms that can guide quantum development in directions that benefit humanity while minimizing

potential risks. This includes developing international standards for quantum technologies, establishing principles for responsible quantum research and development, and creating mechanisms for addressing the social and economic implications of quantum technologies.

Responsible quantum governance requires proactive rather than reactive approaches. Unlike previous technologies where ethical and social considerations were addressed after widespread deployment, the GQA framework emphasizes the importance of embedding ethical considerations into quantum development from the earliest stages. This includes addressing questions of quantum security and cryptography, the implications of quantum computing for privacy and surveillance, and the potential impacts of quantum technologies on employment and social structures.

2.3 Guiding Principles

The Global Quantum Initiative is guided by six fundamental principles that shape all activities and decisions:

Inclusivity: All countries and stakeholders should have opportunities to participate in quantum development according to their needs and capabilities. This principle requires active efforts to overcome barriers to participation and to ensure that diverse voices are heard in quantum governance discussions.

Responsibility: Quantum development should be guided by ethical considerations and social responsibility, with careful attention to potential risks and unintended consequences. This includes transparency in research and development processes, accountability for the impacts of quantum technologies, and commitment to using quantum capabilities for beneficial purposes. This also includes acknowledging lessons from past science and technology booms where early promise led to inflated expectations without parallel investment in enabling environments, ultimately eroding trust and support.

Sustainability: Quantum initiatives should be designed for long-term impact rather than short-term visibility, with attention to environmental sustainability, economic viability, and social sustainability. This principle emphasizes the importance of building lasting capabilities rather than creating dependencies.

Cooperation: Quantum challenges are global in scope and require collaborative solutions that transcend national, sectoral, and disciplinary boundaries. This principle emphasizes the importance of multilateral approaches and shared responsibility for quantum governance.

Transparency: Quantum governance processes should be open and transparent, with clear accountability mechanisms and regular reporting on progress and challenges. This includes transparency in decision-making processes, resource allocation, and evaluation of outcomes.

Adaptability: The quantum landscape is rapidly evolving, requiring governance frameworks that can adapt to new developments and changing circumstances. This principle emphasizes

the importance of learning from experience and adjusting approaches based on evidence and feedback.

3. Implementation Framework: A Phased Approach to Sustainable Impact

3.1 Strategic Rationale for Phased Implementation

The Global Quantum Initiative adopts a deliberately phased approach to implementation, **recognizing the complexity of quantum governance challenges and the importance of building sustainable foundations before scaling up activities**. This approach directly addresses the credibility risks identified in other international initiatives by ensuring that each phase delivers concrete, measurable outcomes that build confidence and momentum for subsequent phases.

The three-year Global Quantum Initiative is the essential preparatory phase (2026-2028) serves multiple strategic purposes: it allows UNESCO and partners to test and validate key concepts and approaches through pilot projects before committing to larger-scale implementation; it provides time to build genuine partnerships and secure sustainable funding commitments; it enables iterative development and refinement of the GQA framework based on practical experience and stakeholder feedback; and it establishes UNESCO and the partners' credibility and leadership in quantum governance through demonstrated results rather than ambitious promises.

This phased approach also aligns with the natural timeline of quantum technology development. Many quantum technologies are still in early research stages, providing a window of opportunity to establish governance frameworks before widespread deployment. By 2030, if a formal GQA is envisioned to be launched, the two-year interval (2029-2030) preceding the envisioned launch of the formal GQA is a dedicated consolidation and commitment period. This stage is pivotal to translating the proof-of-concept and lessons from the GQI into broad-based political ownership. It will focus on securing the formal endorsement and sustained commitment of Member States and key partners, fostering a shared understanding of the GQA's necessity, and finalizing its common mechanisms, working methods, and governance structure.

3.2 Phase 1: Foundation Building (2026)

The first phase of GQI focuses on establishing the institutional foundations necessary for effective quantum governance. This includes developing the conceptual framework for the GQA, building initial partnerships, and launching pilot activities that demonstrate the value of UNESCO's approach.

Key Activities:

Consensus Building and Framework Development: The phase begins with a **comprehensive consultation process involving member states, scientific communities, and other stakeholders** to refine the GQA framework and build consensus around its objectives and principles. This includes regional consultations in Africa, Latin America and the Caribbean, Asia-Pacific, and other regions to ensure that diverse perspectives are incorporated into the framework development process.

The consultation process will be structured around specific thematic areas rather than general discussions, focusing on practical questions such as: *What are the most pressing quantum capacity building needs in different regions? How can international cooperation in quantum research be structured to benefit all participants? Should ethical principles guide quantum development? How can quantum education be made accessible and relevant in different cultural contexts?*

Partnership Development: Phase 1 emphasizes **building strategic partnerships with key stakeholders who can contribute to GQA implementation**. This includes formal partnerships with other international organizations, leading research institutions, and private sector entities committed to responsible quantum development.

Partnership agreements will specify concrete contributions and commitments rather than general expressions of support. For example, research institutions might commit to providing remote access to quantum facilities for researchers from developing countries, while private sector partners might contribute to quantum education initiatives or provide technical expertise for capacity building programs.

Pilot Project Launch: The phase includes the launch of carefully selected pilot projects that demonstrate key elements of the GQA approach. These pilots are designed to be small-scale and manageable while providing valuable learning experiences and concrete benefits to participants.

Priority pilot projects include: a quantum education initiative in three African countries, focusing on teacher training and curriculum development; a remote access programme connecting researchers in Latin America with quantum computing facilities in North America and Europe; and a policy dialogue series bringing together quantum policymakers from different regions to share experiences and develop common approaches.

Institutional Capacity Building: Phase 1 includes investments in building capacity of different institutions in Member States to manage quantum governance activities effectively. This includes training staff, developing technical expertise, and establishing systems for project management, monitoring, and evaluation.

3.3 Phase 2: Expansion and Validation (2027)

The second phase builds on the foundations established in Phase 1 by expanding successful pilot projects, launching new initiatives, and beginning the formal drafting process for the GQA document.

Key Activities:

Pilot Project Expansion: Successful pilot projects from Phase 1 are expanded to reach more countries and participants. This expansion is guided by lessons learned from initial implementation and feedback from participants. The expansion process includes systematic documentation of best practices and development of replication guidelines that can be used by other organizations and countries.

Zero Draft Development: Phase 2 includes the **development of a "Zero Draft" of the Global Quantum Agenda document**. This draft synthesizes insights from the consultation process, pilot project experiences, and ongoing research into quantum governance challenges. The Zero Draft serves as a concrete foundation for further consultation and refinement in Phase 3.

The drafting process is designed to be inclusive and transparent, with multiple opportunities for stakeholder input and feedback. Draft sections are circulated for comment as they are developed, and regular briefings are provided to member states and other stakeholders on the drafting progress.

Mechanism Prototyping: Phase 2 includes the development and testing of key mechanisms that will support GQA implementation. This includes prototyping a **Global Quantum Observatory** for tracking quantum development trends and capacity building needs, testing funding mechanisms for quantum cooperation projects, and developing recognition programs for quantum champions and innovators.

These mechanisms are tested on a small-scale during Phase 2 to identify potential challenges and refine operational procedures before full-scale implementation.

Regional Hub Development: Phase 2 includes the establishment of **regional quantum cooperation hubs that can serve as focal points for GQA activities in different parts of the world**. These hubs are designed to be sustainable and locally owned, with UNESCO providing initial support and technical assistance but not ongoing operational funding.

3.4 Phase 3: Consolidation and Launch Preparation (2028)

The third phase focuses on consolidating achievements from the first two phases, finalizing the GQA document, and preparing for the formal launch of the Global Quantum Agenda in 2030.

Key Activities:

GQA Finalization: The *Zero Draft* developed in Phase 2 is refined through additional consultation rounds and negotiation processes involving member states and other stakeholders. The finalization process includes formal review by UNESCO's governing bodies and endorsement by member states.

Mechanism Institutionalization: The mechanisms prototyped in Phase 2 are formalized and prepared for full-scale operation. This includes establishing governance structures, securing sustainable funding, and developing operational procedures.

Impact Assessment and Documentation: Phase 3 includes comprehensive assessment of the impact achieved during the preparatory phase, including evaluation of pilot projects, analysis of capacity building outcomes, and documentation of lessons learned. This assessment serves both accountability purposes and provides valuable guidance for future GQA implementation.

Launch Preparation: The phase concludes with preparation for the formal launch of the Global Quantum Agenda in 2030. This includes planning for a high-level launch event, developing communication strategies, and securing commitments from partners for GQA implementation.

3.5 Monitoring and Evaluation Framework

The implementation of the Global Quantum Initiative (GQI) includes a robust monitoring and evaluation (M&E) mechanism designed to ensure accountability, facilitate learning, and enable adaptive management throughout the three-year process. The M&E framework is built around three levels of indicators, carefully distinguishing between the process of the GQI and the enduring impact of the GQA it produces.

Output Indicators measure the direct products of *GQI activities*, such as the development of GQA, the number of people trained, policies developed, or partnerships established. These indicators provide immediate feedback on implementation progress and help identify operational challenges.

Outcome Indicators measure the **intermediate effects of GQI activities**, such as improvements in quantum readiness, increased international cooperation, or enhanced policy frameworks. These indicators require longer time horizons to measure but provide important insights into the effectiveness of different approaches.

Impact Indicators measure the long-term success and adoption of the Global Quantum Agenda (GQA) itself. These indicators assess the ultimate goal: whether the framework produced by the GQI is effective and influential in the real world, such as examples: Number of countries adopting national quantum strategies aligned with GQA principles; evidence of new international R&D partnerships formed citing the GQA as a basis; reference to the GQA in international standard-setting bodies; qualitative evidence of the GQA being used as a normative framework to guide responsible quantum innovation.

The M&E framework includes both quantitative and qualitative indicators, recognizing that some important outcomes (such as improved trust and cooperation) are difficult to measure numerically but are nonetheless crucial for GQA success.

4. Detailed Budget and Resource Mobilization Strategy

4.1 Total Budget Overview

The Global Quantum Initiative (2026-2028) requires a total budget of **\$3.6 million over three years**. This budget is designed to be realistic and achievable while providing sufficient resources to deliver meaningful outcomes and establish UNESCO's credibility in quantum governance.

The budget allocation reflects strategic priorities and the phased implementation approach:

- **Year 1 (2026):** \$1,066,600 - Foundation building and pilot project launch
- **Year 2 (2027):** \$1,466,600 - Expansion of activities and mechanism development
- **Year 3 (2028):** \$1,066,600 - Consolidation and launch preparation

4.2 Detailed Budget Breakdown by Component

Component 1: Capacity Building and Education (\$1,120,000 – 33.7%)

This component represents the largest budget allocation, reflecting the central importance of capacity building in addressing the quantum divide.

Quantum Education Initiatives (\$480,000): Development of quantum education extra curricula; such as online courses, and teacher training programmes in partnership with educational institutions in Africa, Latin America, and Asia. This includes educational tools' development, teacher training workshops, educational resource creation, and pilot implementation in selected countries.

Policy Maker Training (\$320,000): Regional training programmes for government officials and policy makers on quantum governance, including workshops, study tours, and development of policy toolkits. This component addresses the critical need for informed policy making in quantum development.

Research Capacity Building (\$320,000): Support for early-career researchers from developing countries to participate in quantum research through fellowships, exchange programmes, and remote access to quantum facilities. This includes partnership agreements with leading quantum research institutions.

Component 2: International Cooperation and Partnerships (\$840,000 – 24.1%)

This component supports the development of sustainable cooperation mechanisms and partnerships that will form the backbone of the GQA.

Partnership Development (\$280,000): Costs associated with developing and maintaining strategic partnerships with international organizations, research institutions, and private sector entities. This includes partnership negotiation, legal agreements, and coordination activities.

Regional Consultations and Dialogues (\$350,000): Organization of regional consultation meetings, thematic dialogues, and expert workshops to build consensus around the GQA framework. This includes venue costs, participant travel, interpretation services, and documentation.

Mechanism Development (\$210,000): Development and testing of key GQA mechanisms, including *the Global Quantum Observatory prototype*, funding mechanisms, and recognition programs. This includes technical development, pilot testing, and refinement based on user feedback.

Component 3: Knowledge Products and Communication (\$560,000 – 16.9%)

This component ensures that **GQI activities** generate lasting knowledge products and maintain effective communication with stakeholders.

Research and Analysis (\$240,000): Commissioned research on quantum governance challenges, analysis of global quantum trends, and development of policy recommendations. This includes external consultancies, data collection, and analysis activities.

Publication and Dissemination (\$200,000): Production and dissemination of GQI knowledge products, including policy briefs, technical reports, educational materials, and the Zero Draft of the GQA document. This includes editing, translation, design, and distribution costs.

Communication and Outreach (\$120,000): Development and implementation of communication strategies to raise awareness of GQI activities and engage stakeholders. This includes website development, social media management, and participation in relevant conferences and events.

Component 4: Programme Management and Administration (\$800,000 – 25.3%)

This component covers the essential management and administrative costs required for effective program implementation.

Staff Costs (\$700,000): Salary support for UNESCO two staff (p2 level and a service contract g4 level) dedicated to GQA implementation, including programme coordination, technical expertise, and administrative support.

Operational Costs (\$100,000): General operational expenses including office supplies, communications, travel for UNESCO staff, and other administrative costs necessary for program implementation.

4.3 Resource Mobilization Strategy

The resource mobilization strategy is designed to diversify funding sources and build sustainable support for GQI activities. The strategy includes multiple funding streams:

International Organizations/ NGOs (\$840,000 – 23.3%): UNESCO's core contribution demonstrates organizational commitment and provides a stable foundation for GQI activities. This funding covers essential program management costs and ensures continuity of activities.

Member State Voluntary Contributions (\$1,120,000 - 31%): Targeted fundraising from member states with strong quantum programs or particular interest in quantum governance. Priority targets include countries with significant quantum investments (US, EU, China, Japan, Canada, Australia) and countries that have expressed strong support for inclusive quantum development.

Private Sector Partnerships (\$560,000 – 15.6%): Contributions from private sector entities committed to responsible quantum development. This includes both financial contributions and in-kind support such as access to quantum facilities, technical expertise, and educational resources.

Other International Organizations (\$840,000 – 23.3%): Co-funding from partner organizations such as other UN agencies, International Organisations, regional development banks, and foundations with interests in science and technology development.

The resource mobilization strategy emphasizes the importance of securing multi-year commitments rather than annual funding, providing greater predictability and enabling more effective program planning.

4.4 Cost-Effectiveness and Value Proposition

The total three-year budget of \$3.6 million is less than the annual budget of many national quantum programs and significantly less than the cost of a single quantum computer system.

The cost-effectiveness of the GQI approach derives from several factors:

Leverage of Existing Infrastructure: The initiative builds on UNESCO's existing institutional infrastructure, field offices, and partner networks, avoiding the need to create new institutions from scratch.

Partnership Multiplication: Strategic partnerships multiply the impact of UNESCO's investment by leveraging partner resources and capabilities. For example, partnerships with research institutions provide access to quantum facilities worth millions of dollars.

Focus on Capacity Building: Investment in human capacity and institutional development creates lasting benefits that continue beyond the initial funding period, providing sustainable returns on investment.

Preventive Approach: Addressing quantum governance challenges proactively is significantly more cost-effective than attempting to address problems after they have become entrenched.

The value proposition extends beyond immediate financial returns to include strategic benefits for UNESCO: establishment of leadership in an emerging and important field, strengthening of relationships with member states and partners, enhancement of UNESCO's reputation for innovation and relevance, and positioning for future funding opportunities in quantum governance.

5. Risk Assessment and Mitigation Strategies

5.1 Organizational Credibility Risks

Risk: Overpromising and Underdelivering

The most significant risk to UNESCO's credibility **is the possibility of announcing ambitious goals that cannot be achieved with available resources**. This risk is particularly acute given both the high-profile nature of quantum technologies and historical precedents in other frontier fields, where early momentum, as in the case of graphene, was not sustained due to overextension without ecosystem readiness.

Mitigation Strategy: The GQI is explicitly designed with conservative goal-setting and realistic timelines. All objectives are linked to specific, measurable outcomes with clear success criteria. The phased implementation approach allows for course correction if initial assumptions prove incorrect. Regular monitoring and transparent reporting ensure that any challenges are identified and addressed promptly.

Risk: Political Interference in Technical Processes

Quantum technologies **have significant national security implications, creating risks that political considerations** could interfere with technical and scientific processes within the GQA framework.

Mitigation Strategy: The GQI framework includes clear separation between technical and political processes, with technical decisions made by expert committees and political oversight limited to high-level strategic direction. Transparency in decision-making processes and clear criteria for technical decisions help prevent political interference.

Risk: Perception of Bias Toward Particular Countries or Regions

Given the geopolitical sensitivity of quantum technologies, there is a risk that the GQA could be perceived as biased toward particular countries or regions, undermining its legitimacy and effectiveness.

Mitigation Strategy: The GQI emphasizes inclusive participation and transparent decision-making processes. Regional balance is built into governance structures, and all major decisions require consensus among diverse stakeholders. Regular independent evaluation helps identify and address any actual or perceived biases.

5.2 Implementation Risks

Risk: Insufficient Partner Commitment

The success of the GQI depends heavily on **sustained commitment** from partners, including member states, research institutions, and private sector entities. There is a risk that initial enthusiasm may not translate into long-term commitment.

Mitigation Strategy: Partnership agreements include specific commitments and accountability mechanisms. The phased implementation approach allows partners to demonstrate commitment gradually rather than requiring large upfront commitments. Regular partner meetings and communication help maintain engagement and address concerns promptly.

Risk: Technological Obsolescence

The rapid pace of quantum technology development creates risk that GQI activities could become obsolete or irrelevant as technologies evolve.

Mitigation Strategy: The GQI is designed to be adaptive and responsive to technological developments. Regular technology scanning and expert consultation help identify emerging trends and adjust activities accordingly. Focus on fundamental capacity building and governance principles rather than specific technologies provides resilience against technological change.

Risk: Competition from Other Initiatives

The proliferation of quantum initiatives creates a risk that the GQI could be overshadowed by better-funded or more visible initiatives, reducing its impact and relevance.

Mitigation Strategy: The GQI emphasizes UNESCO's unique value proposition in inclusion and capacity building, areas where other initiatives have limited focus. Strategic partnerships with other initiatives help avoid duplication and create synergies. Clear communication of GQA's distinctive contributions helps maintain visibility and relevance.

5.3 Financial Risks

Risk: Funding Shortfalls

There is a risk that the full budget may not be mobilized, requiring reduction in activities or extension of timelines.

Mitigation Strategy: The budget is structured with core and optional components, allowing for scaled implementation if full funding is not available. Multiple funding sources reduce dependence on any single donor. Early fundraising efforts and multi-year commitments provide greater financial predictability.

Risk: Currency Fluctuations and Inflation

Multi-year budgets are subject to risks from currency fluctuations and inflation, particularly given the international scope of GQI activities.

Mitigation Strategy: Budget planning includes contingency reserves for currency and inflation risks. Where possible, costs are denominated in stable currencies or indexed to inflation. Regular budget reviews allow for adjustments based on changing economic conditions.

5.4 Stakeholder Risks

Risk: Limited Developing Country Participation

Despite the focus on inclusion, there is a risk that developing countries may lack the capacity or resources to participate meaningfully in GQI activities.

Mitigation Strategy: The GQI includes specific support mechanisms for developing country participation, including travel funding, technical assistance, and capacity building. Activities are designed to be accessible to countries with different levels of quantum readiness. Regional hubs provide local support and reduce participation barriers.

Risk: Private Sector Disengagement

Private sector engagement is crucial for GQI success, but there is a risk that commercial interests may not align with GQI objectives over time.

Mitigation Strategy: Private sector partnerships are structured around shared value creation rather than pure philanthropy. Clear value propositions for private sector partners help maintain engagement. Diverse private sector partnerships reduce dependence on any single company or sector.

6. Strategic Announcement and Launch Framework

6.1 Positioning Within the Global Quantum Ecosystem

The announcement of **the Global Quantum Initiative requires careful positioning to distinguish UNESCO's initiative from existing quantum programs while demonstrating complementarity rather than competition**. The current quantum landscape includes numerous national initiatives, private sector programs, and international collaborations, each with different focuses and constituencies.

The GQI's unique positioning emphasizes three key differentiators: **inclusive development** that prioritizes Global South participation rather than technological competition; **holistic governance** that addresses social, ethical, and educational dimensions alongside technical development; and **multilateral legitimacy** through UNESCO's established role as a neutral convener of diverse stakeholders.

6.2 Announcement Timeline and Milestones

The announcement strategy follows a carefully sequenced timeline designed to build momentum and stakeholder engagement:

Phase 1: Internal Preparation (Q3 2025)

- Finalization of proposal and budget - Internal UNESCO approval processes - Initial stakeholder consultations with key member states - Development of communication materials and messaging

Phase 2: Strategic Pre-Announcement (Q4 2025)

- Briefings for key member states and partners - Engagement with quantum community leaders and influencers - Preparation of launch event and supporting activities - Media preparation and stakeholder outreach

Phase 3: Formal Announcement (IYQ Closing Ceremony, February 2026)

- High-level launch event during IYQ closing ceremony - Director-General announcement and member state endorsements - Media launch and global communication campaign - Immediate follow-up with stakeholder engagement activities

Phase 4: Implementation Launch (Q3 2026)

- Formal commencement of GQI activities - First partner agreements and funding commitments - Launch of initial pilot projects and consultation processes - Establishment of governance structures and operational procedures

6.3 Key Messages and Communication Strategy

The communication strategy emphasizes three core messages that resonate with different stakeholder groups:

For Member States: *"UNESCO's Global Quantum Agenda ensures that all countries can participate in and benefit from the quantum revolution, strengthening national sovereignty and development opportunities while fostering international cooperation."*

For Scientific Community: *"The GQA provides a trusted platform for international quantum cooperation that respects scientific integrity while addressing the social and ethical implications of quantum development."*

For Civil Society: *"UNESCO's leadership ensures that quantum technologies are developed responsibly and inclusively, with attention to human rights, social equity, and sustainable development."*

The communication strategy utilizes multiple channels and formats to reach diverse audiences: high-level diplomatic communications for government stakeholders, technical briefings for scientific communities, accessible summaries for media and civil society, and targeted outreach for specific regions and constituencies.

6.4 Stakeholder Engagement and Coalition Building

The announcement strategy includes systematic engagement with key stakeholder groups to build support and participation:

Member State Engagement: Bilateral consultations with key member states, including major quantum-investing countries and developing countries that would benefit from GQI activities. Regional group briefings through UNESCO's established diplomatic channels. Integration of GQA discussions into relevant UNESCO governing body meetings.

Scientific Community Engagement: Presentations at major quantum conferences and scientific meetings. Engagement with quantum professional societies and research networks. Collaboration with leading quantum researchers and institutions to develop technical content and validate approaches.

Private Sector Engagement: Targeted outreach to quantum companies and technology firms interested in responsible development. Engagement with industry associations and standards organizations. Development of value propositions that align business interests with GQI objectives.

Civil Society Engagement: Collaboration with organizations focused on science policy, technology governance, and international development. Engagement with educational institutions and teacher networks. Outreach to youth organizations and student groups interested in quantum science.

6.5 Managing Expectations and Building Credibility

The announcement strategy includes specific measures to manage expectations and build credibility from the outset:

Realistic Goal Setting: All public communications emphasize the realistic, achievable nature of GQI objectives and the phased implementation approach. Specific timelines and milestones are provided to enable accountability and progress tracking.

Transparency in Resource Requirements: The budget and resource mobilization strategy are communicated transparently, demonstrating that the initiative is properly planned and resourced. The relationship between funding levels and expected outcomes is clearly articulated.

Evidence-Based Approach: The announcement includes reference to UNESCO's preliminary quantum survey results and other evidence supporting the need for the GQA. Pilot project plans are presented as learning opportunities rather than guaranteed successes.

Partnership Emphasis: The collaborative nature of the GQI and GQA is emphasized, making clear that success depends on partner contributions and that UNESCO is facilitating rather than controlling quantum development.

7. Conclusion and Recommendations

7.1 Strategic Imperative for Action

The quantum revolution presents **both unprecedented opportunities and significant risks** for global development and international cooperation. The emergence of a "*quantum divide*" between quantum-advanced and quantum-excluded countries threatens to replicate and amplify existing inequalities in the digital age. Without deliberate intervention, quantum technologies risk becoming another source of technological dependence and geopolitical tension rather than a tool for shared prosperity and sustainable development.

UNESCO's unique mandate and capabilities position the organization to play a crucial leadership role in shaping the global quantum future. The proposed Global Quantum Agenda provides a realistic, achievable framework for addressing quantum governance challenges while building UNESCO's credibility and influence in an emerging field of critical importance.

The timing is optimal for UNESCO's engagement. Quantum technologies are still in early development stages, providing a window of opportunity to establish governance frameworks before widespread deployment. The momentum generated by the International Year of Quantum Science and Technology creates favorable conditions for launching new international cooperation initiatives. The growing recognition of quantum governance challenges among policymakers and researchers creates demand for the kind of neutral, inclusive platform that UNESCO can provide.

7.2 Balancing Ambition with Realism

The proposed framework deliberately balances the ambition vision for global leadership with the realistic constraints of available resources and organizational capacity. The "**Global Quantum Agenda**" name satisfies the branding while the phased implementation approach of the "**Global Quantum Initiative**" and conservative goal-setting protect against credibility risks.

This balance is achieved through several design features: realistic budget requirements that represent a significant but manageable investment for UNESCO; phased implementation of GQI that allows for learning and adaptation; emphasis on capacity building and cooperation rather than technological competition; focus on UNESCO's comparative advantages in inclusion and multilateral cooperation; and built-in accountability mechanisms that ensure transparency and enable course correction.

The framework addresses the quantum divide concern through targeted capacity building, South-South cooperation mechanisms, and emphasis on equitable participation. It satisfies the need for uniqueness by focusing on governance and inclusion dimensions that are underemphasized in existing initiatives. It protects organizational credibility through realistic goal-setting, transparent resource allocation, and robust monitoring and evaluation systems.

7.3 Implementation Recommendations

Immediate Actions (Q3-Q4 2025):

- 1. Secure Internal Approval:** Present this proposal to UNESCO's senior management for approval and endorsement. Ensure alignment with organizational priorities and resource allocation decisions.
- 2. Begin Stakeholder Engagement:** Initiate consultations with key member states, particularly those with significant quantum investments and those that would benefit most from GQA activities. Build an initial coalition of supporters.

3. Develop Partnership Pipeline: Begin discussions with potential partners, including international organizations, research institutions, and private sector entities. Develop preliminary partnership agreements and contribution commitments.

4. Prepare Launch Event: Plan high-level launch event for IQ closing ceremony, including speaker recruitment, venue arrangements, and media strategy.

Short-term Actions (Q1-Q2 2026):

1. Establish Governance Structure: Create GQI steering committee and technical advisory groups with appropriate regional and stakeholder representation. Develop operational procedures and decision-making processes.

2. Launch Pilot Projects: Begin implementation of carefully selected pilot projects that demonstrate GQI value and generate early wins. Focus on activities with high visibility and clear benefits.

3. Secure Initial Funding: Complete fundraising for Year 1 activities and secure multi-year commitments where possible. Diversify funding sources to reduce dependence on any single donor.

4. Begin Consultation Process: Launch comprehensive consultation process with member states and stakeholders to develop and refine GQA framework and build consensus around implementation approaches.

Medium-term Actions (2026-2028):

1. Scale Successful Activities: Expand pilot projects that demonstrate success and discontinue or modify those that face challenges. Use evidence-based approach to guide scaling decisions.

2. Develop GQA Document: Complete drafting process for formal GQA document through inclusive consultation and negotiation processes. Ensure broad stakeholder buy-in and member state endorsement.

3. Institutionalize Mechanisms: Establish sustainable governance and operational mechanisms that can continue beyond the preparatory phase. Secure long-term funding and institutional commitments.

4. Prepare for Formal Launch: Plan comprehensive launch strategy for formal GQA in 2030, including high-level political endorsement and operational readiness.

7.4 Success Metrics and Accountability

The success of the Global Quantum Agenda will be measured against specific, measurable indicators that reflect both quantitative achievements and qualitative impacts:

Quantitative Indicators:

- Number of countries actively participating in GQI activities (**target: 50+ by 2028**)
- Number of individuals trained through GQI capacity building programs (**target: 1,000+ by 2028**)
- Number of partnerships established and maintained (**target: 25+ by 2028**)
- Amount of co-funding mobilized from partners (**target: \$2M+ by 2028**)
- Number of policy frameworks influenced or developed (**target: 15+ by 2028**)

Qualitative Indicators:

- Improvement in quantum readiness in participating developing countries
- Enhanced international cooperation and trust in quantum governance
- Increased awareness and understanding of quantum technologies among policymakers
- Development of ethical frameworks and responsible governance practices
- Strengthened UNESCO leadership and credibility in emerging technology governance

Regular monitoring and evaluation will track progress against these indicators, with annual reports providing transparent accountability to member states and partners. Independent evaluation will assess both implementation effectiveness and strategic impact, providing guidance for adaptive management and future planning.

7.5 Final Recommendations

The Global Quantum Initiative represents a strategic opportunity for UNESCO to establish leadership in an emerging field of critical importance while addressing fundamental challenges of equity and inclusion in technological development. The proposed Global Quantum Agenda framework balances ambition with realism, providing a pathway for meaningful impact.

The quantum revolution is not inevitable in its current trajectory. With thoughtful intervention and inclusive governance, quantum technologies can become tools for shared prosperity and sustainable development rather than sources of division and inequality. UNESCO's Global Quantum Agenda provides a roadmap for achieving this vision through practical, achievable steps that build on the organization's unique strengths and capabilities.

The time for action is now. The quantum future is still being written, and UNESCO has the opportunity to help ensure that it is a future that benefits all of humanity.

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The Global Quantum Agenda: Ensuring quantum technologies serve humanity's highest aspirations for knowledge, cooperation, and shared prosperity.

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